Sai Vikram Adithya Bokka
San Jose State University
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CAREER OVERVIEW

5+ years of expertise in block-level PNR, STA, PPA trade-offs, ensuring timing closure, addressing Physical Verification challenges, and performing ECOs for signoff.

EDUCATION

Master of Science in Electrical Engineering, VLSI, 3.5/4.0 Aug 2022 - May 2024
San Jose State University, San Jose, CA

- Course Work: Adv Digital System Design and Synthesis, Adv Computer Architecture, ASIC CMOS, ML, Semiconductor Devices, Computer Vision.

B. Tech, Electronics and Communication Engineering, 3.5/4.0 Jun 2014 - Jun 2018
Jawaharlal Nehru Technology University, Hyderabad, Telangana, India

- Received "Best Alumni Award" for the best outcome of the year 2018.

SKILSS

Professional Tools: Synopsys - DSO-AI, ICC2, Fusion Compiler, ICC2, DC, FEV, Prime Time; Cadence - Innovus, Genus; Redhawk, Conformal, Formality, LT Spice, Visio.
Scripting Languages: TCL, Perl, Python and Shell Scripting.
Expertise: Block Level PnR, Synthesis, Floor Planning, Placement, CTS, Routing, ECO's, LVS, DRC, ERC, Timing Caliber, Timing Closure, PPA, Layout Verification, IR-RV, FEV, STA, TCL Scripting.

WORK EXPERIENCE

Graduate Physical Design Intern, Intel Corporation, San Jose, California, USA May 2023 - Dec 2023

- Experimented with the latest Design Space Optimization (DSO.Ai) AI wrapper from Synopsys uses the AI/ML Algorithms for a better PPA, QoR's and design trade off's and worked on a complex block fastening timing, DRV's and PV.
- Understanding and functioning on ERC fixes, timemod, checking timing arcs, Full Chip STA, and IR-RV - **Intel FPGA Project**.

SoC Design Engineer, Intel Technology India Pvt Limited, Bangalore, India Sep 2020 - Aug 2022

- Recognized multiple times in intel for going beyond the deliverables and exceeding the expectations.
- Controlled two block-level partitions, one with 3.6M transistors and the other with 500K RC partition, both running at 2 GHz, involving tasks all PNR activities and taped out **TSMC 3nm – Intel SoC Project**.
- Achieved a block-level partition with ~3.3M transistors, operating at a clock frequency of 1.8 GHz, containing a high RP count block with 128 RP's, did multiple iterations to sign off on timing, ECOs, and Physical Verification **TSMC 5nm – Intel SoC Project.**
- Led Full Chip timing calibration, collaborating with RTL and PNR owners, as well as TFM teams, to ensure Full Chip timing convergence with no violations **TSMC 5nm – Meteor Lake**.
- Computerized multiple tasks, creating reports and to save time by authored using TCL language.

Physical Design Engineer, Laksh Semiconductors Pvt Limited, Bangalore, India Dec 2018 - Aug 2020

- Managed the largest and most complex block in the project, with ~4.6M gates 707 MHz, involving multiple floorplan iterations with 30 macros and manual placement to optimize (QoR's) and achieve a block utilization of 72.3% **Alder Lake IPU 10+++ nm.**
- Excelled in closing timing, FEV, IR/RV, DRV, DRC's and manual fixes on IR violations.
- Handled multiple partitions for the closure of Unclocked Registers and Unconstrained Endpoints **DG-2 512 – 10+++ nm.**
- Block level partition with 90K instance count and its operating frequency of 1Ghz.
- Signed off on closing block level timing, LVS, DRC and ECO's and cleaned LVS-DRC for multiple blocks **TSMC 7nm project**.

Trainee Engineer, Entuple Technology Pvt Limited, Bangalore, India Feb 2018 - Dec 2018

- Trained in ASIC Physical Design and Verification. Worked on 45nm Advanced Encryption Standard (AES) Project.

ACADEMIC PROJECTS

AI4I2020-Predictive-Maintance: ML Project, San Jose State University Apr 2023

- Built Machine Learning Classification Models using KNN, Logistic regression, Decision Tree, Random Forest, and Confusion Matrix, Cluster Analysis, Feature Selection, Data Modeling, Accuracy, Precision.

Implementation of a 29-tap Symmetrical FIR Filter -ASIC CMOS, San Jose State University Apr 2023

- Designed a 29-tap FIR filter for IQ signals leveraging a limited number of multipliers and optimized for a clock frequency of 300 MHz.
- Achieved a fast filter operation time of 3 clock cycles (100 MHz sample rate), resulting in high-quality fixed binary point 8.24 output data using Synopsys Primetime and Synopsys Design Ware.

Design and Implementation of a 5-stage Pipelined MIPS Processor using Verilog (Computer Architecture), Nov 2022

- Designed and implemented 5 stages of pipelining for the complete execution of an instruction.
- Implemented the RTL Design of the RISC processor using Verilog HDL on Model-sim.